Highlights

The Boussinesq–compressible gap in steady two-dimensional buoyancy-driven enclosure convection is linear to leading order in the relative temperature contrast

- Field error below 1% can coexist with $> 2\%$ Nu error in 43% of cavity samples
- Wall-flux loss reduces matched six-point error against a non-converged 64^2 target
- A two-seed 4-by-4 matrix puts every wall-flux minimum on its training stencil
- A 134-family bootstrap leaves magnitude and temperature-curvature sign unresolved
- Contrast scaling and 3D tests are consistency checks, not general validation